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OpenCV

Initial Activities

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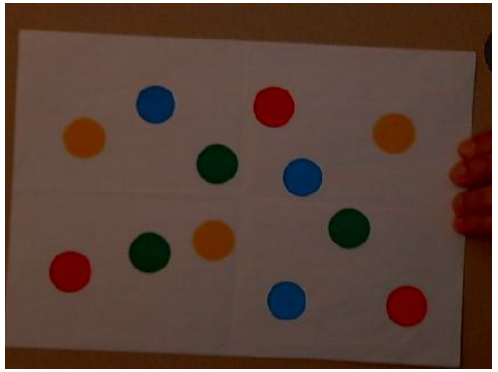
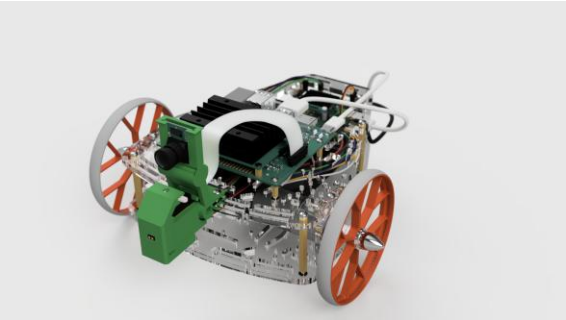
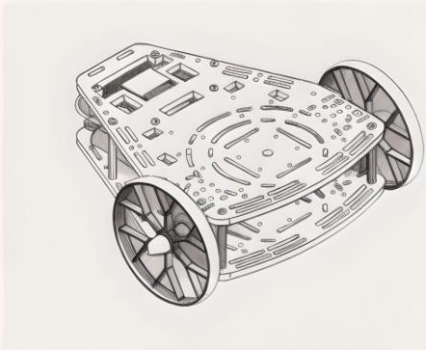
OpenCV Basics

- For the following examples, download the following images and video from GitHub
  - MCR2_Logo_White
  - MCR2_Logo_Black
  - MCR2_Puzzlebot_Jetson_Ed
  - Puzzlebot_logo_black
  - Puzzlebot_logo_white
  - sample_video.mp4







Activity 1

Open a Saved Image

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Opening an image in OpenCV

- Make a new folder called “opencv_course”
- Make a Python file inside the folder called “opencv_imgRead.py”
- Open your preferred IDE or a notepad.
- Save your file as “.py”
- Make sure the previous images are in the folder “opencv_course/images”
- Copy the following code and launch it from terminal in Windows or Ubuntu respectively:

```
$ python opencv_imgRead.py
```

```
$ python3 opencv_imgRead.py
```

```python
import cv2 as cv

#Read the image in the following path "images/MCR2_Logo_Black.png"
img = cv.imread('images/MCR2_Logo_Black.png’)

#If no image is present, finish the program
if img is None:
    print("Failed to load image")
    exit()

#Show the image on a pop-up window
cv.imshow('Example Image', img)

#Wait for any key to be pressed
cv.waitKey(0)

#Destroy all windows
cv.destroyAllWindows()
```

Activity 2

Open a Recorded Video File

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Opening a recorded video in OpenCV

- Make a Python file inside the “opencv_course” folder called “opencv_videoRead.py”
- Open your preferred IDE or a notepad.
- Save your file as “.py”
- Make sure the “sample_video.mp4” is in the folder “opencv_course/videos”
- Copy the following code and launch it from terminal in Windows or Ubuntu respectively:

```
$ python opencv_videoRead.py
```

```
$ python3 opencv_videoRead.py
```

```python
import cv2 as cv

# Read the video file (replace with your actual video file path)
video_path = 'videos/sample_video.mp4'
cap = cv.VideoCapture(video_path)

# Check if the video was successfully opened
if not cap.isOpened():
    print("Failed to load video")
    exit()

# Loop through each frame of the video
while True:
    # Read a frame
    ret, frame = cap.read()
    # If no frame is returned, the video has ended
    if not ret:
        print("End of video or failed to read frame")
        break
    # Display the frame in a window
    cv.imshow('Video Playback', frame)
    # Wait for 25ms and break if 'q' is pressed (adjust time to control playback speed)
    if cv.waitKey(25) & 0xFF == ord('q'):
        print("Playback interrupted by user")
        break
# Release the video capture object and destroy all windows
cap.release()
cv.destroyAllWindows()
```

Activity 3

Live Camera

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Opening camera in OpenCV

- Make a new Python file called "opencv_videoCapture.py"
- Open your preferred IDE or a notepad.
- Save your file as ".py"
- Copy the following code and launch it from terminal in Windows or Ubuntu respectively:

```
$ python opencv_videoCapture.py
```

```
$ python3 opencv_videoCapture.py
```

```python
import numpy as np
import cv2 as cv

#Change parameter 0, ... to access other camera devices
cap = cv.VideoCapture(0)

if not cap.isOpened():
    print("Cannot open camera")
    exit()

while True:
    # Capture frame-by-frame
    ret, frame = cap.read()

    # if frame is read correctly ret is True
    if not ret:
        print("Can't receive frame (stream end?). Exiting ...")
        break
    # Wait until the user press "q" to exit the program
    if cv.waitKey(1) & 0xFF == ord('q'):
        break

    # change color of the image to grayscale
    gray = cv.cvtColor(frame, cv.COLOR_BGR2GRAY)

    # Display the resulting frame
    cv.imshow('frame', gray)

# When everything done, release the capture
cap.release()
cv.destroyAllWindows()
```

Activity 4

Save Images from a Live Feed

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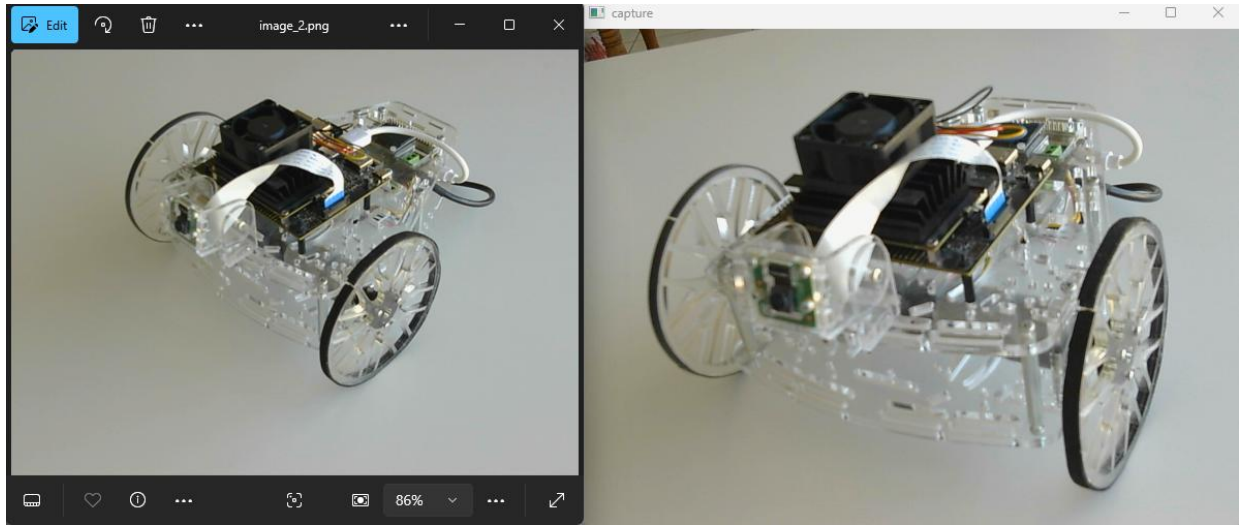
Simple camera image saver

- Make a new Python file called "opencv_saveImages.py"
- Open your preferred IDE or a notepad.
- Save your file as ".py"
- Copy the code on the next slide
- Launch it from terminal in Windows or Ubuntu respectively:

```
$ python opencv_saveImages.py
```

```
$ python3 opencv_saveImages.py
```



Simple camera image saver

```python
# This Python script uses OpenCV to access a webcam, display a live video feed, and capture
# images when the user presses the SPACEBAR.It continuously shows the video stream in a window
# and waits for user input:Press ESC to exit the program.Press SPACEBAR to take a snapshot and
# save it as an image file in the ./images/ directory.

# Import required libraries
import numpy as np  # (Not used in this code, but useful for image processing)
import cv2  # OpenCV library for computer vision tasks

# Initialize the camera (webcam)
# Argument '2' specifies the camera index (can be 0, 1, 2... depending on
available cameras)
cam = cv2.VideoCapture(2)

# Counter for saved images
img_counter = 0

# Start an infinite loop to capture frames from the camera
while True:
    # Read a frame from the camera
    ret, frame = cam.read()

    # If the frame was not successfully captured, exit the loop
    if not ret:
        print("failed to grab frame")
        break

    # Display the current frame in a window titled 'capture'
    cv2.imshow("capture", frame)

    # Wait for a key press for 1 millisecond
    k = cv2.waitKey(1)

    # If the ESC key is pressed (ASCII code 27)
    if k % 256 == 27:
        print("Escape hit, closing...")  # Notify user
        break  # Exit the loop

    # If the SPACE key is pressed (ASCII code 32)
    elif k % 256 == 32:
        # Define the filename with the current counter value
        img_name = "./images/image_{}.png".format(img_counter)

        # Save the current frame to the specified file
        cv2.imwrite(img_name, frame)

        # Notify the user that the image was written
        print("{} written!".format(img_name))

        # Increment the counter for the next image
        img_counter += 1

# Release the camera resource after the loop ends
cam.release()

# Close all OpenCV windows
cv2.destroyAllWindows()
```

Activity 5

Simple Video Recorder

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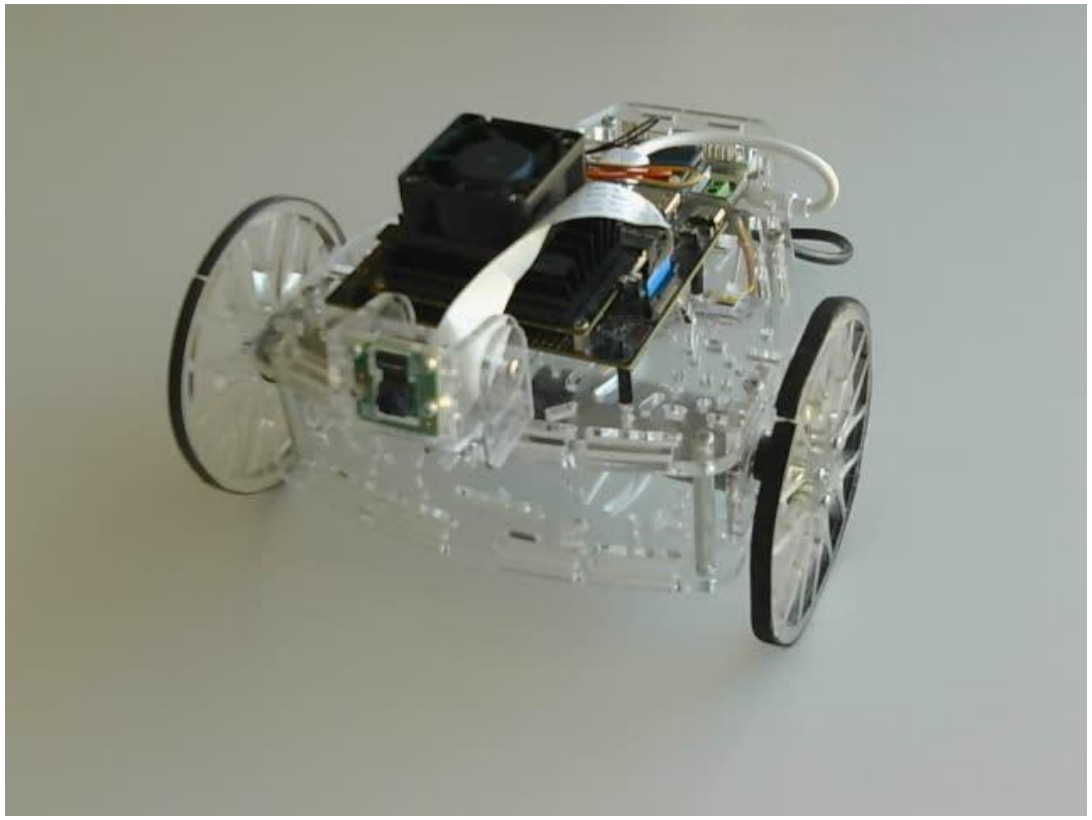
Simple video recorder

- Make a new Python file called "opencv_videoCaptureSave.py"
- Open your preferred IDE or a notepad.
- Save your file as ".py"
- Copy the code on the next slide
- Launch it from terminal in Windows or Ubuntu respectively:

```
$ python opencv_videoCaptureSave.py
```

```
$ python3 opencv_videoCaptureSave.py
```



Simple video recorder

```python
import numpy as np
import cv2 as cv

# Open the camera device (use index 2, can be 0 or 1 depending on your system)
cap = cv.VideoCapture(2)

# Define the codec and prepare to create VideoWriter object later (but don't
start recording yet)
fourcc = cv.VideoWriter_fourcc(*'MJPG')
# Create a new VideoWriter object (you can add timestamp to avoid overwriting)
out = cv.VideoWriter('./videos/Test.avi', fourcc, 30.0, (640, 480))

# Boolean flag to track whether recording is active
recording = False

while cap.isOpened():
    # Read a frame from the camera
    ret, frame = cap.read()

    # If the frame wasn't read successfully, exit the loop
    if not ret:
        print("Can't receive frame (stream end?). Exiting ...")
        break

    # If recording is ON, write the current frame to the video file
    if recording:
        out.write(frame)

    # Display the current frame in a window named 'frame'
    cv.imshow('frame', frame)

    # Listen for keyboard events (check every 1 ms)
    k = cv.waitKey(1) & 0xFF
    # If the 'q' key is pressed, exit the loop
    if k == ord('q'):
        print("Quitting...")
        break

    # If the SPACEBAR is pressed, toggle recording ON/OFF
    elif k == ord(' '):  # SPACE key
        recording = not recording  # Toggle the recording state
        if recording:
            # Start recording: create the VideoWriter object
            print("Recording started...")

        else:
            # Stop recording: release the VideoWriter object
            print("Recording stopped.")
            break

# Release the camera resource when the loop ends
cap.release()
# If recording was still on when quitting, release the writer
out.release()
# Close all OpenCV windows
cv.destroyAllWindows()
```

Activity 5

Simple Pixel Manipulation

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Image manipulation

- Make a new Python file called "opencv_pixelManip.py"
- Open your preferred IDE or a notepad.
- Save your file as ".py"
- Copy the code on the next slide
- Launch it from the terminal in Windows or Ubuntu respectively:

```
$ python opencv_pixelManip.py
```

```
$ python3 opencv_pixelManip.py
```

Image manipulation

```python
import numpy as np
import cv2 as cv

# Load the image
img = cv.imread('images/MCR2_Logo_Black.png')

# Check if the image was loaded successfully
if img is None:
    print("Failed to load image")
    exit()

# Get the height and width of the image
height, width, channels = img.shape

# Calculate the center coordinates
center_y = int(height / 2)
center_x = int(width / 2)

# Set a single pixel at the center to white
img[center_y, center_x] = [255, 255, 255]
```

```python
# Fill a region around the center (e.g., a white rectangle of size 100x100)
# Define the size of the square (half the length from the center)
square_size = 50

# Fill the region with black color
img[0:square_size, width-square_size:width] = [0,0,0]

# Show image
cv.imshow("Display window", img)

# Wait indefinitely until any key is pressed
cv.waitKey(0)

# Close all OpenCV windows
cv.destroyAllWindows()
```

Thank you

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